

Bishop's University

Department of Computer Science

Programming languages for DS - CS504

Final exam - Fall 2023

Problelem

1. Solve the problem of face recognition with Logistic regression using Scikit-learn and Keras/Pytorch

Note:

- Do not create new cells for your codes. Use only those I have created to provide your answers. Otherwise you will get marks deducted.

1. Load libraries required to solve the problem

```
In [1]: # import

import warnings
warnings.filterwarnings('ignore')
```

2. Load face dataset

https://scikit-learn.org/stable/datasets/real_world.html#olivetti-faces-dataset

- Carefully read the dataset description, structure, sample shape, size, etc.

```
In [2]: # Load dataset from the link shown above
```

3. Extract data sample in a matrix X and class labels in a vector y

```
In [3]: # X: samples, y: class labels
```

```
In [6]: print(X.shape, y.shape)
```

```
(400, 4096) (400,)
```

4. Display one random face image

```
In [1]: # Select on row from X randomly, reshape it to a 64*64 matrix then display it as as
```

5. Split the data into training and test datasets

- Keep the input samples as vectors and not reshape them to images

```
In [4]: # split into 70% train and 30% test
```

```
In [10]: print(X_train.shape, X_test.shape)
```

```
(280, 4096) (120, 4096)
```

6. Implemente Logistic Regression using Scikit-learn library

6.1. Train the model on the training set using 10-fold cross-validation startegy and evaluate it on the test set.

```
In [ ]: # training on training set
```

```
# Evaluation on test set
```

6.2. Display the classification report

```
In [ ]: # Classificaion report, f-score, recall and precision, all in a table called classi
```

7. Implemente Logistic Regression using Keras or Pytorch (your choice)

1. Use one input layer for features and one output layer for prediction. The number of nodes in each layer depends on the number of features and the number of classes respectively.
2. Construct the model.
3. Define the optimizer (SGD, ADAM, ...) with its parameters.
4. Compile and fit the model to the training set using 10-fold cross validation strategy.
5. Evaluate the model on the test set and show the classification report.

```
In [ ]: # training on training set
```

```
# evaluation on test set
```

7.2. Add some hidden layers (with different number of nodes each) between the input and the output layers. Train and evaluate the model as in previous question.

```
In [ ]:
```

7.3. Analyse the performance of the model with respect to the number of hidden layers.

a) Using Bar plot, show the accuracy vs the number of hidden layers.

```
In [ ]:
```

8. Analyse the performance of the model with respect to the optimizer

8.1. Use different types of optimizers and show the results using Bar plot.

```
In [ ]:
```

9. Reduce the dimensionality of the problem using PCA.

9.1. Use the 2 first principal component and plot the data scatter of the training set.

```
In [ ]:
```

9.2. Analyse the performance of the Logistic Regression model (using Scikit-learn package) with respect to the number of principal components used, for example (2,3,4,...10). Show the results using Bar plot.

```
In [ ]:
```